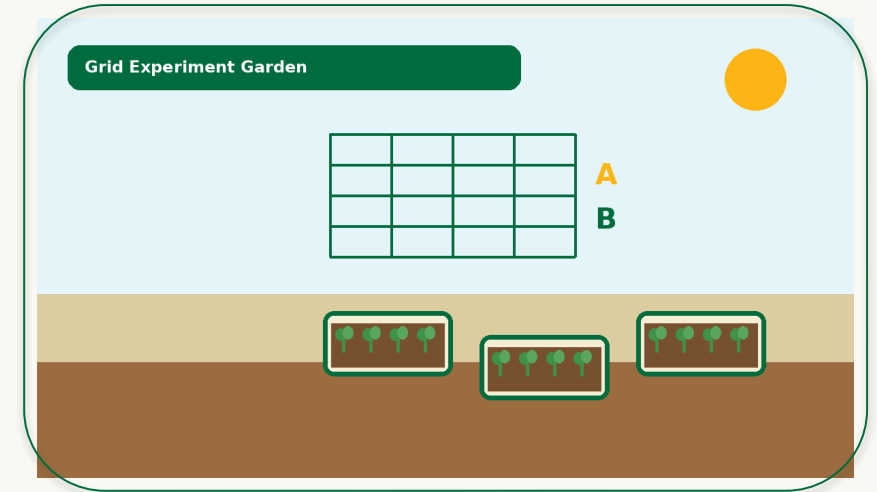


Grid Experiment Garden

Introduction to Agricultural Experiments

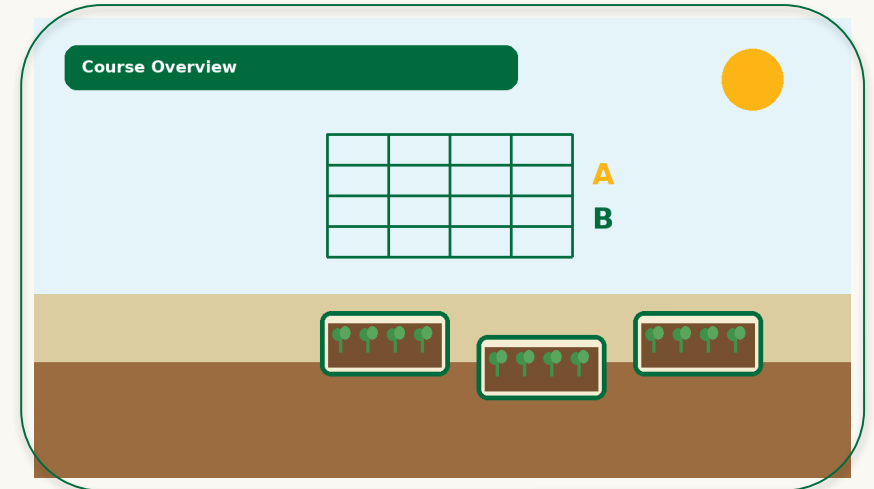
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Course emphasis: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden system and its educational purpose; teach the ten technical topics at Primary 4-6 level; connect mathematics/science/agriculture to practical work; supervise measurements and garden tasks safely; maintain usable records; guide pupils from observation toward evidence-based conclusions.
- Core materials: teacher guide, crop plan, child-safe tools, watering equipment, labels, clipboards, rulers/tapes, scales or thermometers where relevant, observation/data sheets, clean containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson/measurement or investigation plan.



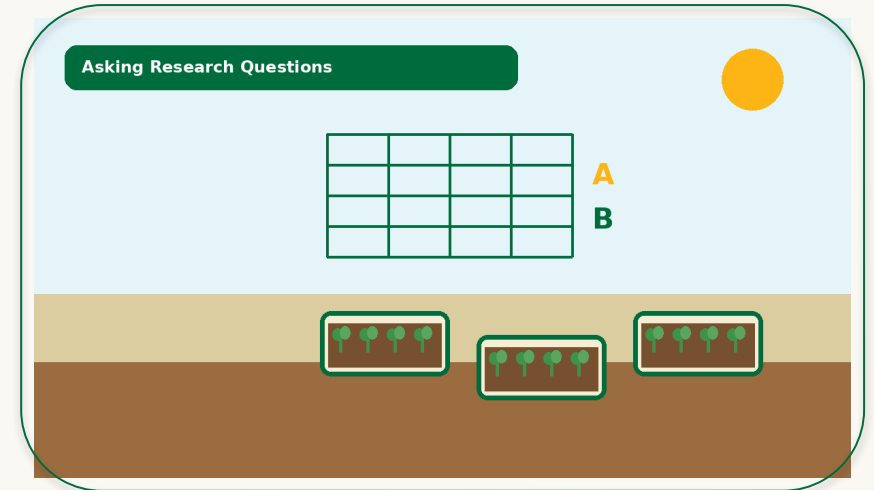
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

Asking Research Questions

Teacher knowledge and practical application

- Start with questions that can be answered by observation or measurement in the school garden.
- Keep one main comparison at a time for beginners.
- Examples: Does mulch change how quickly surface soil dries? Do two varieties differ in growth?
- Avoid questions requiring unsafe treatments or equipment.
- Translate broad curiosity into a specific measurable question.



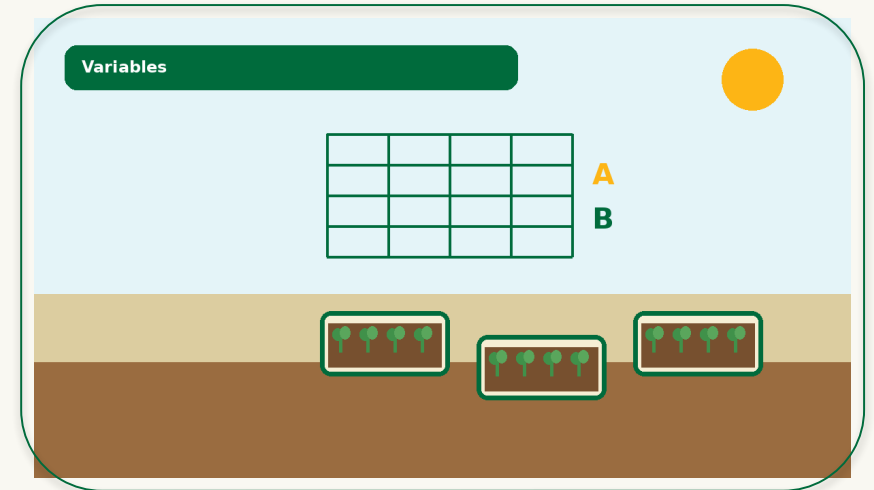
TEACHER PRACTICE

- Write one feasible garden research question for Primary 4-6.

Variables

Teacher knowledge and practical application

- Introduce a variable as something that can change or be measured.
- Distinguish the factor deliberately changed from the response measured.
- Identify conditions teachers try to keep similar.
- Use concrete garden examples rather than abstract terminology alone.
- Do not imply that every uncontrolled difference can be eliminated outdoors.



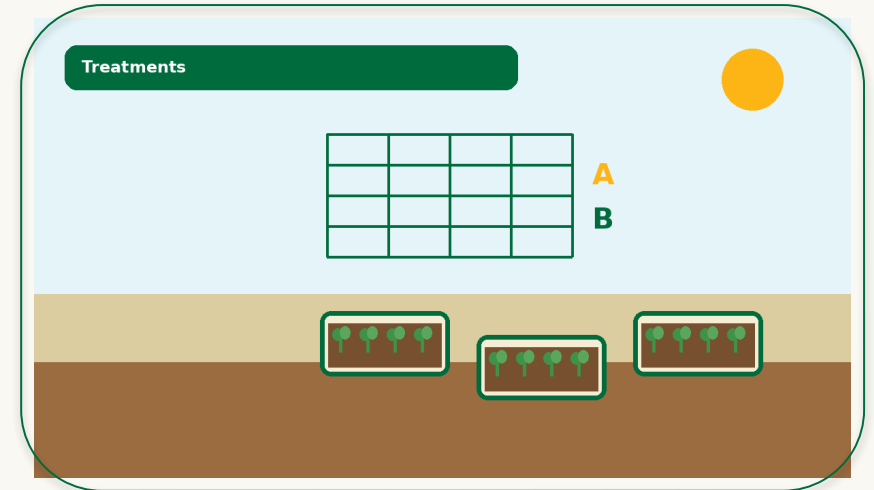
TEACHER PRACTICE

- Identify changed, measured and kept-similar variables in one example.

Treatments

Teacher knowledge and practical application

- Define treatments as the planned conditions being compared.
- Use safe, meaningful treatments such as mulch/no mulch or two approved crop varieties.
- Apply each treatment consistently.
- Label plots clearly before the experiment starts.
- Avoid treatment differences that create food-safety or plant-health risks.



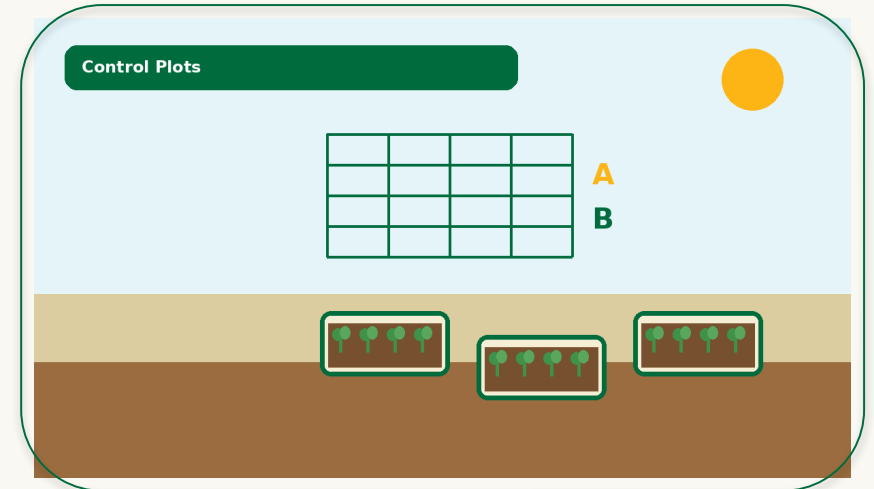
TEACHER PRACTICE

- Design two simple treatments for a class investigation.

Control Plots

Teacher knowledge and practical application

- Explain a control as a reference condition used for comparison when appropriate.
- Use the normal or standard garden practice as the control where that makes sense.
- Treat control plots with the same care except for the factor being tested.
- Label controls clearly.
- Explain that not every school investigation requires a formal control.



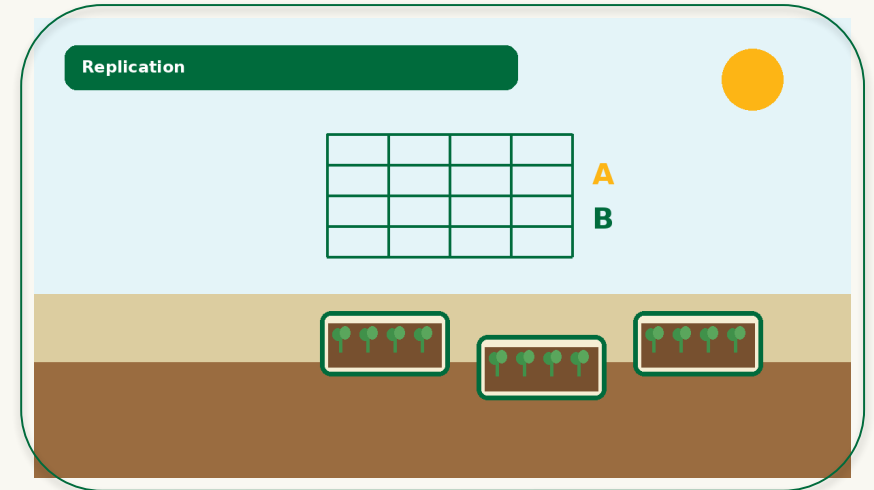
TEACHER PRACTICE

- Identify an appropriate control for the planned investigation.

Replication

Teacher knowledge and practical application

- Explain replication as repeating the same treatment in more than one plot or unit.
- Show how repeated plots help reveal natural variation.
- Use at least simple repeated units when space allows.
- Do not confuse multiple plants within one plot with independent replicated plots without explanation.
- Keep replication manageable for the class.



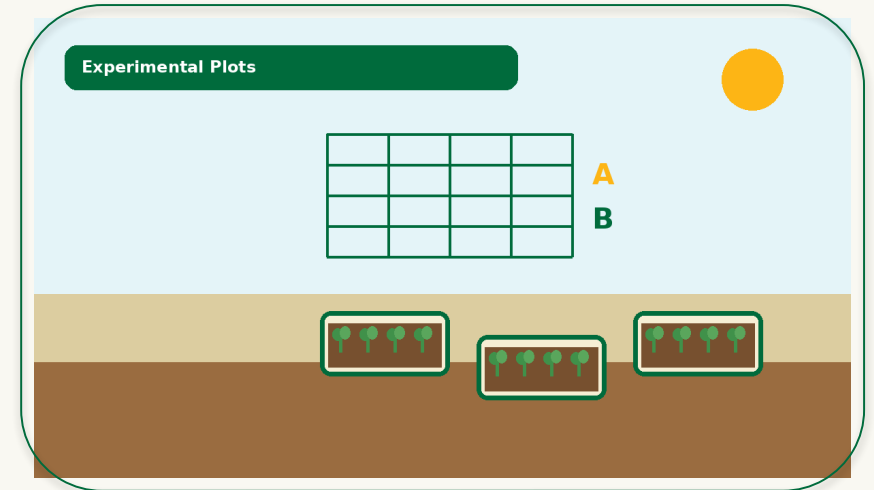
TEACHER PRACTICE

- Sketch a simple replicated two-treatment layout.

Experimental Plots

Teacher knowledge and practical application

- Use a numbered grid or clearly bounded plots.
- Keep plot size and basic management similar across treatments where possible.
- Use buffer space if needed to prevent treatment interference.
- Create a plot map before planting.
- Maintain labels throughout the investigation.



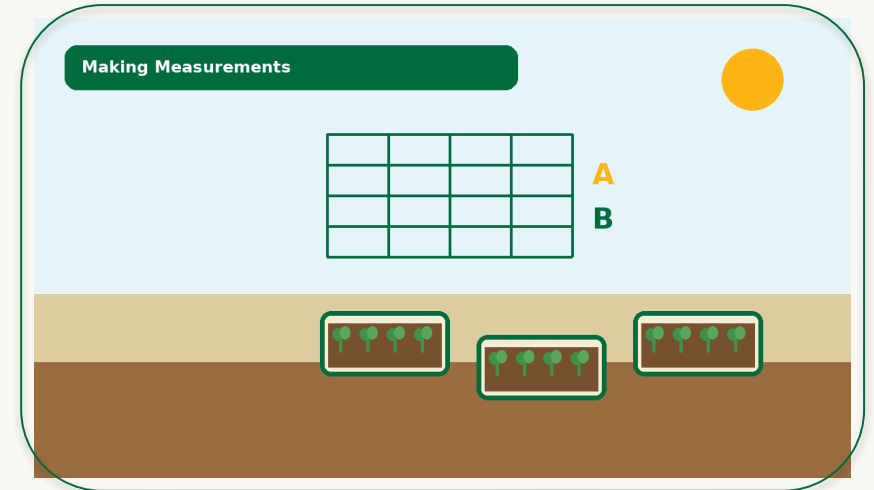
TEACHER PRACTICE

- Prepare a plot map with treatment labels.

Making Measurements

Teacher knowledge and practical application

- Select measurements directly connected to the research question.
- Use the same method, tool, time interval and unit across treatments.
- Mark sampled plants where repeated measurements are required.
- Teach pupils to measure without choosing only the 'best-looking' plants.
- Record unusual events that may affect data.



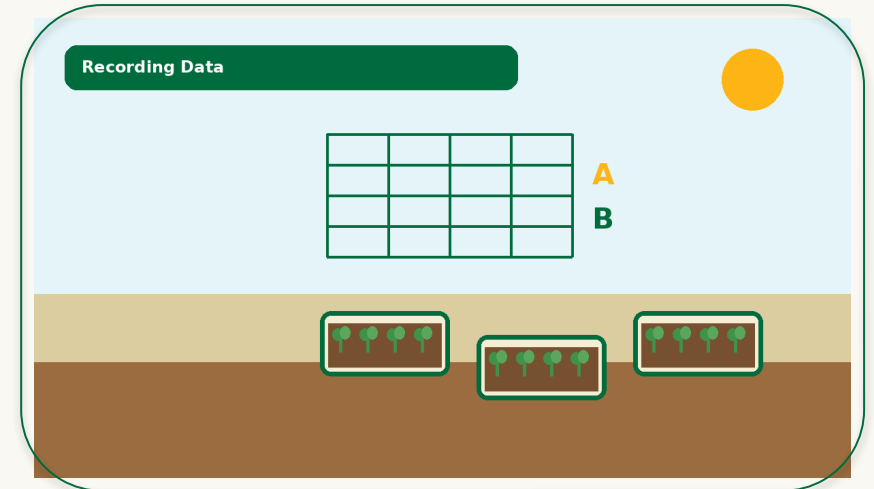
TEACHER PRACTICE

- Write a measurement protocol for one response variable.

Recording Data

Teacher knowledge and practical application

- Use a table prepared before data collection.
- Record raw observations immediately rather than relying on memory.
- Include date, plot/treatment, measurement and unit.
- Do not erase unexpected values simply because they look different.
- Check entries for transcription mistakes.



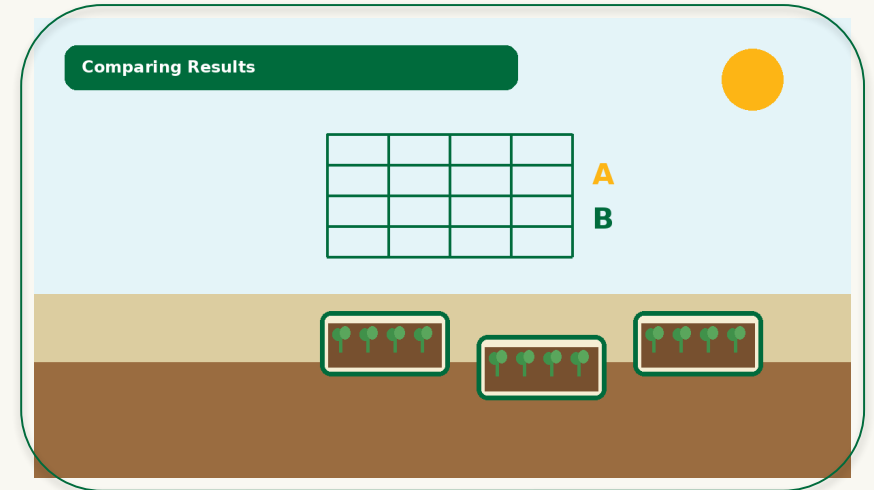
TEACHER PRACTICE

- Create a class data sheet for the experiment.

Comparing Results

Teacher knowledge and practical application

- Summarize results using simple averages, counts or graphs where grade-appropriate.
- Compare treatment groups using the same metric.
- Look for patterns while acknowledging variation.
- Do not claim a cause solely because one treatment value is larger in a small school trial.
- Return to the original research question.



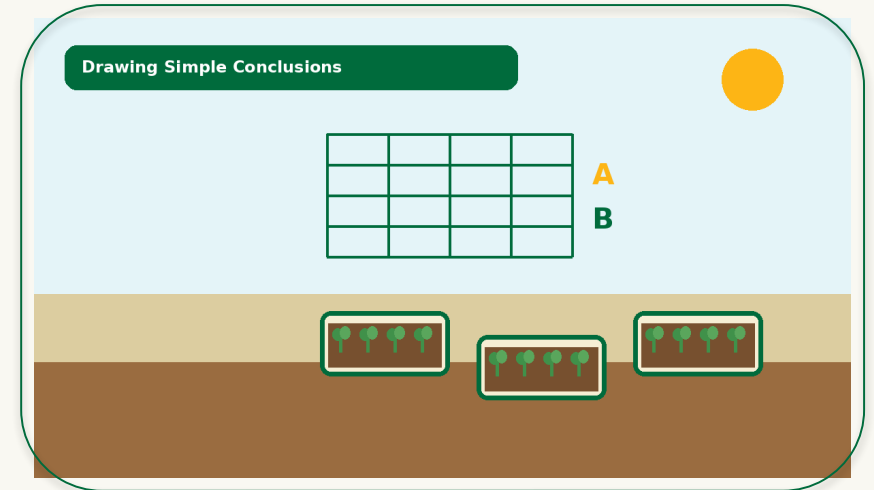
TEACHER PRACTICE

- Prepare one table and one simple graph from sample data.

Drawing Simple Conclusions

Teacher knowledge and practical application

- State what the class observed and whether it supported the initial expectation.
- Keep conclusions limited to the experiment actually conducted.
- Mention important limitations such as small sample size, weather or uneven emergence.
- Suggest what could be tested next.
- Distinguish evidence from opinion.



TEACHER PRACTICE

- Completion task: prepare a question-to-conclusion mini experiment plan.